

Q.1)

Consider a system with 6 processes P1, P2, P3, P4, P5 and P6 who are requesting 4 resource types – A, B, C and D at the same time. The following snapshot of the system has been given at the moment.

| Process | Maximum |   |   |   | Allocation |   |   |   | Available |   |   |   |
|---------|---------|---|---|---|------------|---|---|---|-----------|---|---|---|
|         | A       | B | C | D | A          | B | C | D | A         | B | C | D |
| P1      | 4       | 2 | 1 | 2 | 2          | 0 | 0 | 1 | 3         | 3 | 2 | 1 |
| P2      | 5       | 2 | 5 | 2 | 3          | 1 | 2 | 1 |           |   |   |   |
| P3      | 2       | 3 | 1 | 6 | 2          | 1 | 0 | 3 |           |   |   |   |
| P4      | 1       | 4 | 2 | 4 | 1          | 3 | 1 | 2 |           |   |   |   |
| P5      | 3       | 6 | 6 | 5 | 1          | 4 | 3 | 2 |           |   |   |   |
| P6      | 4       | 2 | 3 | 5 | 1          | 0 | 2 | 2 |           |   |   |   |

A) Compute the Need Matrix for the above data.

(4 points)

B)

Is the system in a safe state? Why? If yes, show the safe sequence. Kindly show each and every step of your solution.

(8 points)

C)

Now, if a request from process P2 arrives for (1,1,0,1), should the request be granted immediately? Why or Why not? Kindly show each and every step of your solution.

(8 points)